Problem F. Gift

Input file:	`standard input`
Output file:	`standard output`
Time limit:	1 second
Memory limit:	256 megabytes

Given an undirected graph with n vertices and n edges, you need to calculate how many ways there are to choose a vertex p and an edge (x, y) such that, after removing the edge (x, y), the graph becomes a tree, and when this tree is rooted at p, each node has no more than 3 children. It is guaranteed that there is at least one possible plan.

Input

The first line of the input contains a single integer n ($2 \le n \le 10^5$).

The next n lines of the input describes the edges of the graph. The i-th line of these lines contains two integers x_i and y_i ($1 \le x_i, y_i \le n$, $x_i \ne y_i$), indicating the i-th edge.

It is guaranteed that there are no multiple edges or self loops in the graph.

Output

Output a single line contains a single integer, indicating the answer.

Example

standard input	standard output
6 1 2 1 3 1 4 1 5 1 6 2 3	10